Probability and Statistics Final Exam Retake Solution

December 18, 2025

---

1 Correlation Test (Test for Dependence)

We test if there is a significant positive correlation between a computer being from Factory A and it breaking down. This is equivalent to testing the independence of the two factors using a Chi-square test.

1. Contingency Table (Observed Counts)

Let A be the event the computer is from Factory A, and B be the event it breaks down. Total computers $n = 100 + 200 = 300$.

	Broken Down (B)	**Working** (B^c)	**Total**
Factory A (A)	$O_{11} = 17$	$100 - 17 = 83$	$n_A = 100$
Factory B (A^c)	$O_{21} = 9$	$200 - 9 = 191$	$n_{A^c} = 200$
Total	$n_B = 26$	$n_{B^c} = 274$	$n = 300$

2. Hypotheses and Expected Count

We test for independence (H_0) against dependence (H_A).

$$H_0 : \text{Factory and Breakdown are independent}$$

$$H_A : \text{Factory and Breakdown are dependent (positive correlation)}$$

The expected count for E_{11} (Factory A and Broken Down) under H_0 is:

$$E_{11} = \frac{n_A \cdot n_B}{n} = \frac{100 \cdot 26}{300} \approx 8.67$$

3. Test Statistic (χ^2 and Decision)

The observed Chi-square statistic is $\chi^2_{\text{obs}} \approx 20.35$ with $\nu = 1$ degree of freedom. The critical value is $\chi^2_{1;0.05} = 3.841$. Since $\chi^2_{\text{obs}} = 20.35 > 3.841$, we **reject** H_0. Furthermore, since the observed count (17) is much greater than the expected count (8.67), the dependence is positive. We conclude there is a **significant positive correlation**.

---

2 Server Response Time (Z-Test for Mean)

We test a hypothesis about the population mean (μ) with a **known population variance** ($\sigma^2 = 10$).

1. Hypotheses (Two-Tailed Test)

$$H_0 : \mu = 300 \quad \text{vs} \quad H_A : \mu \neq 300$$

2. Given Data and Sample Mean

- Sample size: $n = 5$.
- Population standard deviation: $\sigma = \sqrt{10} \approx 3.162$ ms.
- Observations: $245, 341, 287, 269, 302$.
- Sample mean: $\bar{X} = 288.8$ ms.
- Significance level: $\alpha = 0.05$.

3. Test Statistic (Z-statistic)

$$Z_{\text{obs}} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{288.8 - 300}{3.162/\sqrt{5}} \approx -7.92$$

4. Critical Region and Decision

For a two-tailed Z-test at $\alpha = 0.05$, the critical value is $z_{0.025} = 1.96$. Since $|Z_{\text{obs}}| = 7.92$ is much greater than 1.96, we **reject** H_0. We cannot accept that the expectation of the response time is equal to 300 ms.

---

3 Peter's Hypothesis (χ^2 Goodness-of-Fit Test)

We test if the observed distribution of sports activity matches Peter's theoretical distribution.

1. Hypotheses

Peter's hypothesis H_0: $(p_1, p_2, p_3) = (0.30, 0.40, 0.30)$.

$$H_0 : \text{The distribution matches Peter's claim} \quad \text{vs} \quad H_A : H_0 \text{ is false}$$

2. Observed and Expected Counts

Total sample size $n = 300$. Degrees of freedom: $\nu = 2$.

- Observed (O_k): Regular=125, Occasional=130, Never=45.
- Expected (E_k): Regular=90, Occasional=120, Never=90.

3. Test Statistic (χ^2_{obs})

$$\chi^2_{\text{obs}} = \frac{(125-90)^2}{90} + \frac{(130-120)^2}{120} + \frac{(45-90)^2}{90} \approx 36.94$$

4. Critical Region and Decision

For a χ^2 test with $\nu = 2$ d.f. and $\alpha = 0.05$, the critical value is $\chi^2_{2;0.05} = 5.991$. Since $\chi^2_{\text{obs}} = 36.94$ is much greater than 5.991, we **reject** H_0. We cannot accept Peter's hypothesis.

---

4 Unbiased Estimator for Uniform Distribution Parameters

The random variables $X_1, \ldots, X_n$ are $U[3+t, 5+8t]$. We seek a, b such that $E[a\overline{X} + bs_n^{*2}] = t$.

1. Population Parameters in terms of t

- Population Mean: $\mu = 4 + \frac{9}{2}t$.
- Population Variance: $\sigma^2 = \frac{(2+7t)^2}{12}$.

2. Analysis of Unbiasedness

The requirement $E[a\overline{X} + bs_n^{*2}] = t$ leads to:

$$a\left(4 + \frac{9}{2}t\right) + b\left(\frac{4}{12} + \frac{28}{12}t + \frac{49}{12}t^2\right) = t$$

Since the variance (σ^2) contains a t^2 term, and the target parameter (t) does not, no such unbiased linear combination exists. The constraints derived from equating coefficients (specifically the t^2 and constant terms) lead to a contradiction.

Conclusion: The parameter t cannot be estimated unbiasedly by a linear combination of $\overline{X}$ and s_n^{*2}. (The constants a and b do not exist.)

---

5 New COVID Cases (Linear Regression)

We analyze the linear regression model for new COVID cases (y) against the day of November (x).

(a) Goodness of Fit

- $R^2 = 0.958$.

Since 95.8% of the variation in new cases is explained by the linear trend, this is a **very good fit**.

(b) Significance of the Main Coefficient ($\hat{\beta}_1$)

We test $H_0 : \beta_1 = 0$ vs $H_A : \beta_1 \neq 0$ ($\alpha = 0.05$).

- P-value for the day coefficient is 0.004.

Since $P = 0.004 < \alpha = 0.05$, we **reject** H_0. The slope is **significantly different from zero**.

(c) Prediction for November 22, 2021

November 22 is $x = 22$. The estimated regression equation is:

$$\hat{y} = 2888.5000 + 284.4000x$$

Prediction for $x = 22$:

$$\hat{y}_{22} = 2888.5000 + 284.4000(22) = 9145.30 \text{ new cases}$$